

NIGERIAN USE CASES

Amazon DynamoDB

- Use Case: Fintech KYC Document Store
- Why: Flexible schema, unpredictable access patterns, automatic scaling

Amazon RDS (MySQL)

- Use Case: E-commerce Product Catalogue
- Why: Structured product data, SQL queries, relational integrity

Amazon Aurora

- Use Case: Payment Ledger
- Why: ACID transactions, high availability, high throughput

Amazon ElastiCache (Redis)

- Use Case: OTP (One-Time Password) Cache
- Why: Sub-millisecond reads/writes, temporary data storage

Amazon Redshift

- Use Case: MTN/Airtel Subscriber Analytics
- Why: Petabyte-scale analytics, reporting, trend analysis

MANAGED RDS vs MYSQL ON EC2

Task	Amazon RDS	MySQL on EC2
OS Patching	AWS	You
Database Patching	AWS	You
Backups	AWS Automated	You Configure
Failover	AWS Multi-AZ	You Configure
Monitoring	AWS CloudWatch	You Configure
Scaling	Easier	Manual
High Availability	Built-in Options	Custom Setup
Maintenance Effort	Low	High

Summary:

- Choose Amazon RDS when you want AWS to handle operational tasks.
- Choose MySQL on EC2 when you need full OS/database control and custom configurations.

Amazon QLDB (Quantum Ledger Database) for Payment Audit Trails

- **Immutable Transaction History:** QLDB provides a cryptographically verifiable, append-only ledger where records cannot be altered or deleted. For companies like Remita or Interswitch, this creates a trusted audit trail for every payment transaction, helping prevent fraud and unauthorized modifications.
- **Simplified Compliance and Auditing:** Financial institutions and payment processors must demonstrate transaction integrity to regulators and auditors. QLDB automatically maintains a complete history of all changes, making regulatory reporting, dispute resolution, and compliance audits much easier than with traditional databases.
- **Built-in Data Verification:** QLDB uses cryptographic hashing to verify that transaction records have not been tampered with. This allows payment companies to prove the authenticity and integrity of their transaction history without implementing complex custom ledger solutions.